Service to Product in 60 Days

The DSA + System Design calendar that got me into a product company.

60	45	15	240
days	days of DSA	days of System Design	tasks to tick

How to use this planner

- 4 tasks a day. Each one has a checkbox: print the pages and tick them off.
- Days 1 to 45: DSA, 4 LeetCode problems per day, from arrays to DP, ending with mock rounds.
- Days 46 to 60: System Design, 4 topics per day, from fundamentals to full designs and mocks.
- Solve for 2 hours, then write down 1 learning. Revise starred problems on weekends.
- Purple cards are DSA days. Teal cards are System Design days.

Days 1 to 12

Arrays, Strings, Searching, Linked Lists

DSA

DAY 01 — ARRAYS

- [ ] Two Sum
- [ ] Best Time to Buy and Sell Stock
- [ ] Contains Duplicate
- [ ] Product of Array Except Self

DAY 02 — ARRAYS

- [ ] Maximum Subarray
- [ ] Merge Intervals
- [ ] Rotate Array
- [ ] Move Zeroes

DAY 03 — ARRAYS

- [ ] 3Sum
- [ ] Container With Most Water
- [ ] Next Permutation
- [ ] Majority Element

DAY 04 — STRINGS

- [ ] Valid Anagram
- [ ] Longest Substring Without Repeating Characters
- [ ] Group Anagrams
- [ ] Valid Palindrome

DAY 05 — STRINGS

- [ ] Longest Palindromic Substring
- [ ] String to Integer (atoi)
- [ ] Longest Common Prefix
- [ ] Minimum Window Substring

DAY 06 — HASHING

- [ ] Subarray Sum Equals K
- [ ] Longest Consecutive Sequence
- [ ] Top K Frequent Elements
- [ ] LRU Cache

DAY 07 — TWO POINTERS / WINDOW

- [ ] Trapping Rain Water
- [ ] Sort Colors
- [ ] Permutation in String
- [ ] Max Consecutive Ones III

DAY 08 — BINARY SEARCH

- [ ] Binary Search
- [ ] Search in Rotated Sorted Array
- [ ] Find Minimum in Rotated Sorted Array
- [ ] Koko Eating Bananas

DAY 09 — BINARY SEARCH

- [ ] Median of Two Sorted Arrays
- [ ] Search a 2D Matrix
- [ ] Capacity to Ship Packages
- [ ] Find Peak Element

DAY 10 — LINKED LIST

- [ ] Reverse Linked List
- [ ] Merge Two Sorted Lists
- [ ] Linked List Cycle
- [ ] Remove Nth Node From End

DAY 11 — LINKED LIST

- [ ] Reorder List
- [ ] Add Two Numbers
- [ ] Copy List with Random Pointer
- [ ] Merge K Sorted Lists

DAY 12 — STACK

- [ ] Valid Parentheses
- [ ] Min Stack
- [ ] Daily Temperatures
- [ ] Largest Rectangle in Histogram

Tick each box when done. Missed a day? Move it forward, never skip it.

Days 13 to 24 Stacks, Backtracking, Trees, Heaps, Graphs

DAY 13 — STACK / QUEUE

- [ ] Next Greater Element I
- [ ] Evaluate Reverse Polish Notation
- [ ] Implement Queue using Stacks
- [ ] Sliding Window Maximum

DAY 14 — BACKTRACKING

- [ ] Subsets
- [ ] Permutations
- [ ] Combination Sum
- [ ] Letter Combinations of a Phone Number

DAY 15 — BACKTRACKING

- [ ] Word Search
- [ ] N-Queens
- [ ] Palindrome Partitioning
- [ ] Generate Parentheses

DAY 16 — TREES

- [ ] Maximum Depth of Binary Tree
- [ ] Invert Binary Tree
- [ ] Same Tree
- [ ] Diameter of Binary Tree

DAY 17 — TREES (BFS)

- [ ] Binary Tree Level Order Traversal
- [ ] Zigzag Level Order Traversal
- [ ] Binary Tree Right Side View
- [ ] Lowest Common Ancestor of a Binary Tree

DAY 18 — BST

- [ ] Validate Binary Search Tree
- [ ] Kth Smallest Element in a BST
- [ ] LCA of a BST
- [ ] Convert Sorted Array to BST

DAY 19 — TREES (HARD)

- [ ] Construct Tree from Preorder and Inorder
- [ ] Serialize and Deserialize Binary Tree
- [ ] Binary Tree Maximum Path Sum
- [ ] Subtree of Another Tree

DAY 20 — HEAPS

- [ ] Kth Largest Element in an Array
- [ ] Top K Frequent Words
- [ ] Find Median from Data Stream
- [ ] Task Scheduler

DAY 21 — HEAPS

- [ ] Last Stone Weight
- [ ] K Closest Points to Origin
- [ ] Reorganize String
- [ ] Meeting Rooms II

DAY 22 — GRAPHS (DFS/BFS)

- [ ] Number of Islands
- [ ] Clone Graph
- [ ] Max Area of Island
- [ ] Rotting Oranges

DAY 23 — GRAPHS (GRID)

- [ ] Pacific Atlantic Water Flow
- [ ] Surrounded Regions
- [ ] Walls and Gates
- [ ] Word Ladder

DAY 24 — GRAPHS (TOPO SORT)

- [ ] Course Schedule
- [ ] Course Schedule II
- [ ] Alien Dictionary
- [ ] Graph Valid Tree

Tick each box when done. Missed a day? Move it forward, never skip it.

Days 25 to 36 Graphs, Tries, Dynamic Programming, Greedy

DAY 25 — UNION FIND

- [ ] Number of Connected Components
- [ ] Redundant Connection
- [ ] Accounts Merge
- [ ] Is Graph Bipartite?

DAY 26 — SHORTEST PATHS

- [ ] Network Delay Time
- [ ] Cheapest Flights Within K Stops
- [ ] Path With Minimum Effort
- [ ] Min Cost to Connect All Points

DAY 27 — TRIES

- [ ] Implement Trie
- [ ] Design Add and Search Words
- [ ] Word Search II
- [ ] Replace Words

DAY 28 — DP (1D)

- [ ] Climbing Stairs
- [ ] House Robber
- [ ] House Robber II
- [ ] Min Cost Climbing Stairs

DAY 29 — DP (1D)

- [ ] Coin Change
- [ ] Longest Increasing Subsequence
- [ ] Word Break
- [ ] Decode Ways

DAY 30 — DP (SUBSET)

- [ ] Maximum Product Subarray
- [ ] Partition Equal Subset Sum
- [ ] Combination Sum IV
- [ ] Perfect Squares

DAY 31 — DP (2D GRID)

- [ ] Unique Paths
- [ ] Minimum Path Sum
- [ ] Maximal Square
- [ ] Triangle

DAY 32 — DP (STRINGS)

- [ ] Longest Common Subsequence
- [ ] Edit Distance
- [ ] Palindromic Substrings
- [ ] Distinct Subsequences

DAY 33 — DP (HARD)

- [ ] Best Time to Buy/Sell Stock with Cooldown
- [ ] Target Sum
- [ ] Regular Expression Matching
- [ ] Burst Balloons

DAY 34 — GREEDY

- [ ] Jump Game
- [ ] Jump Game II
- [ ] Gas Station
- [ ] Hand of Straights

DAY 35 — INTERVALS

- [ ] Non-overlapping Intervals
- [ ] Insert Interval
- [ ] Meeting Rooms
- [ ] Partition Labels

DAY 36 — BIT MANIPULATION

- [ ] Single Number
- [ ] Number of 1 Bits
- [ ] Counting Bits
- [ ] Missing Number

Tick each box when done. Missed a day? Move it forward, never skip it.

Days 37 to 48 Revision, Mocks, then System Design begins

DSA + SD

DAY 37 — BITS / MATH

- [ ] Reverse Bits
- [ ] Sum of Two Integers
- [ ] Pow(x, n)
- [ ] Happy Number

DAY 38 — MATRIX

- [ ] Rotate Image
- [ ] Spiral Matrix
- [ ] Set Matrix Zeroes
- [ ] Game of Life

DAY 39 — DESIGN DS

- [ ] Design HashMap
- [ ] Design Twitter
- [ ] Time Based Key-Value Store
- [ ] Insert Delete GetRandom O(1)

DAY 40 — REVISION: ARRAYS

- [ ] First Missing Positive
- [ ] Find the Duplicate Number
- [ ] Longest Repeating Character Replacement
- [ ] Minimum Size Subarray Sum

DAY 41 — Revision: Trees/Graphs

- [ ] Flatten Binary Tree to Linked List
- [ ] Populating Next Right Pointers
- [ ] Count Good Nodes in Binary Tree
- [ ] Reconstruct Itinerary

DAY 42 — REVISION: DP

- [ ] Longest Palindromic Subsequence
- [ ] House Robber III
- [ ] Interleaving String
- [ ] Coin Change II

DAY 43 — MOCK ROUND 1

- [ ] Medium problem #1 (30 min, timed)
- [ ] Medium problem #2 (30 min, timed)
- [ ] Hard problem (45 min, timed)
- [ ] Review mistakes and note patterns

DAY 44 — MOCK ROUND 2

- [ ] Medium problem #1 (30 min, timed)
- [ ] Medium problem #2 (30 min, timed)
- [ ] Hard problem (45 min, timed)
- [ ] Explain solutions out loud

DAY 45 — FINAL DSA REVISION

- [ ] Re-solve your 4 toughest problems
- [ ] Revise pattern notes (window, DP, graphs)
- [ ] Complexity cheat-sheet review
- [ ] Dry-run 2 problems on paper

DAY 46 — SD: FUNDAMENTALS

- [ ] Scalability: vertical vs horizontal
- [ ] Latency vs throughput
- [ ] CAP theorem and consistency models
- [ ] Load balancing (L4 vs L7)

DAY 47 — SD: CACHING

- [ ] Cache strategies (write-through, write-back, aside)
- [ ] Eviction policies (LRU, LFU)
- [ ] CDN basics
- [ ] Redis internals and use cases

DAY 48 — SD: DATABASES

- [ ] SQL vs NoSQL: when to use which
- [ ] Indexing and B-Trees
- [ ] Replication (leader-follower)
- [ ] Sharding and partitioning

Tick each box when done. Missed a day? Move it forward, never skip it.

Days 49 to 60 Classic Designs, LLD, Reliability, Mocks

DAY 49 — SD: APIS AND NETWORK

- [ ] REST vs gRPC vs GraphQL
- [ ] HTTP/HTTPS, DNS, TCP basics
- [ ] WebSockets, SSE, long polling
- [ ] Rate limiting algorithms

DAY 50 — SD: MESSAGING

- [ ] Message queues vs streams
- [ ] Kafka fundamentals
- [ ] Pub/Sub and event-driven design
- [ ] Idempotency, retries, dead-letter queues

DAY 51 — SD: BUILDING BLOCKS

- [ ] Consistent hashing
- [ ] Bloom filters
- [ ] Unique ID generation (Snowflake)
- [ ] Back-of-envelope estimation

DAY 52 — DESIGN: CLASSICS

- [ ] URL Shortener (TinyURL)
- [ ] Pastebin
- [ ] Rate Limiter
- [ ] Unique ID Generator

DAY 53 — DESIGN: INFRA

- [ ] Key-Value Store
- [ ] Notification System
- [ ] Web Crawler
- [ ] Typeahead / Autocomplete

DAY 54 — DESIGN: SOCIAL

- [ ] WhatsApp (Chat System)
- [ ] Twitter News Feed
- [ ] Instagram
- [ ] Dropbox / Google Drive

DAY 55 — DESIGN: LARGE SCALE

- [ ] YouTube / Netflix
- [ ] Uber / Ride Sharing
- [ ] Google Docs (Collaboration)
- [ ] BookMyShow / Ticketmaster

DAY 56 — LLD

- [ ] SOLID principles
- [ ] Design patterns (Singleton, Factory, Observer)
- [ ] Parking Lot (LLD)
- [ ] Splitwise / Elevator (LLD)

DAY 57 — RELIABILITY

- [ ] Monolith vs microservices
- [ ] API gateway and service discovery
- [ ] Circuit breaker, bulkhead, retries
- [ ] Monitoring, logging, tracing

DAY 58 — MOCK DESIGN 1

- [ ] Pick a random problem, 45 min timer
- [ ] Requirements, estimates, API design
- [ ] High-level diagram and deep dive
- [ ] Self-review: gaps and trade-offs

DAY 59 — MOCK DESIGN 2

- [ ] Pick a new problem, 45 min timer
- [ ] Explain trade-offs out loud
- [ ] Handle follow-ups: scale 10x, failures
- [ ] Record yourself and review

DAY 60 — LAUNCH DAY

- [ ] Revise system design cheat-sheet
- [ ] Prepare 5 STAR behavioural stories
- [ ] Update resume and LinkedIn
- [ ] Ask for 10 referrals and apply

Resources

DSA Practice

- NeetCode 150 and Blind 75
- Striver's A2Z / SDE Sheet
- LeetCode (topic tags + company-wise)
- GeeksforGeeks for concept revision

DSA Learning

- NeetCode YouTube explanations
- Striver (take U forward) playlists
- Abdul Bari: Algorithms
- Cracking the Coding Interview

System Design

- ByteByteGo (Alex Xu books, newsletter)
- System Design Primer (GitHub)
- Gaurav Sen and Hello Interview (YouTube)
- Designing Data-Intensive Applications

LLD and Interview Prep

- Refactoring Guru (design patterns)
- Concept & Coding (LLD playlists)
- Pramp / Interviewing.io for mock interviews
- Referrals via LinkedIn and Naukri